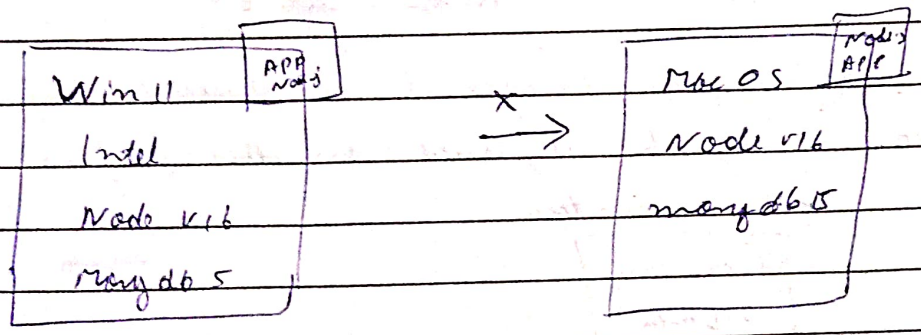
Docker

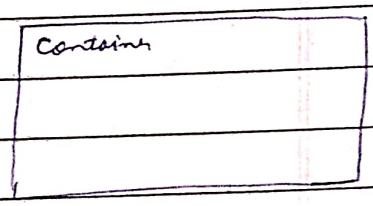
(static, read-only) Blueprint → Image → Container ← (dynamic) running instance of image

Why Docker :

To Replicate the previous Enviroment so that a application can run smoothly on various platforms & services like / including clouds - AWS.



⇒ Container to solve this problem
↳ lightweight, isolated env running on Kernel

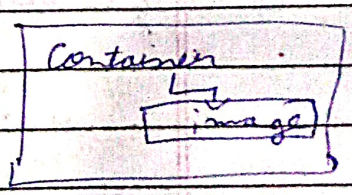


- Daemon

- Docker is Containerisation platform. It allows to package all the dependencies (OS libraries, runtimes, config) into container image.

- Docker desktop → GUI to manage & see Docker

• Run Ubuntu on Mac OS Remo -



⇒ docker -v ∴ version

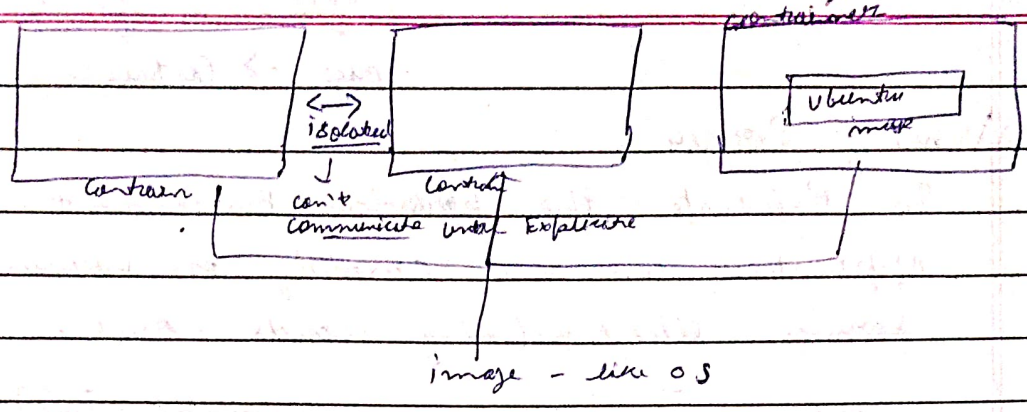
⇒ docker run -it ubuntu ⇐ we goes into ~~ubuntu~~ container where ubuntu image is running.
↑ <image-name>

-it → interactive

Do not let the docker container exit

↳ it check if their is already a image present then download if not it download it from hub.docker.com.

→ docker hub → kind of Github for Containers



- we can make custom images & can publish them on Docker hub → can be deployed on other platforms through it's

eg - ubuntu - with - tool

Custom img
- Ubuntu
- Node
- mongodb
- Redis

- docker container ls ← shows all running containers

- docker container ls -a ← shows all including exited containers
  (or)
  docker ps -a

- docker start Container-name
- docker stop Container-name

- docker exec <container name> <command>

eg :- docker exec mj ls

- docker exec -it <container-name><command>

- docker images
- docker image ls

← to list the images

- docker run -it node